

LAB 10: Hadoop – Spark

Write a simple streaming program in Spark to receive text data streams on a particular port, perform basic text cleaning (like white space removal, stop words removal, lemmatization, etc.), and print the cleaned text on the screen.

Code:

```
# Install NLTK and download required data (run once)

!pip install nltk

import nltk
nltk.download('punkt')
nltk.download('stopwords')
nltk.download('wordnet')

from pyspark.sql import SparkSession
from pyspark.sql.functions import col, lower, regexp_replace, split, explode, udf
from pyspark.sql.types import ArrayType, StringType
from pyspark.ml.feature import StopWordsRemover
from nltk.stem import WordNetLemmatizer

# Initialize SparkSession
spark = SparkSession.builder.appName("TextProcessing").getOrCreate()

# Define your input lines
lines = [
    "Hello, I hate you.",
    "I hate that I love you.",
    "Don't want to, but I can't put",
    "nobody else above you."
]

# Create DataFrame from lines
df = spark.createDataFrame(lines, "string").toDF("value")

# Step 1: Lowercase and remove punctuation
df_clean = df.select(regexp_replace(lower(col("value")), "[^a-zA-Z\\s]", "").alias("cleaned"))

# Step 2: Tokenize the cleaned text
df_tokens = df_clean.select(split(col("cleaned"), "\\s+").alias("tokens"))

# Step 3: Remove stop words
remover = StopWordsRemover(inputCol="tokens", outputCol="filtered")
df_filtered = remover.transform(df_tokens)
```

```
# Step 4: Lemmatization using NLTK WordNetLemmatizer with UDF lemmatizer =  
WordNetLemmatizer()
```

```
def lemmatize_words(words):  
    return [lemmatizer.lemmatize(word) for word in words]
```

```
lemmatize_udf = udf(lemmatize_words, ArrayType(StringType()))
```

```
df_lemmatized = df_filtered.withColumn("lemmatized", lemmatize_udf(col("filtered")))
```

```
# Step 5: Explode the lemmatized words and show results
```

```
df_lemmatized.select(explode(col("lemmatized")).alias("word")).show(truncate=False)
```

```
Requirement already satisfied: nltk in /usr/local/lib/python3.11/dist-packages (3.9.1)  
Requirement already satisfied: click in /usr/local/lib/python3.11/dist-packages (from nltk) (8.2.0)  
Requirement already satisfied: joblib in /usr/local/lib/python3.11/dist-packages (from nltk) (1.5.0)  
Requirement already satisfied: regex>=2021.8.3 in /usr/local/lib/python3.11/dist-packages (from nltk) (2024.11.6)  
Requirement already satisfied: tqdm in /usr/local/lib/python3.11/dist-packages (from nltk) (4.67.1)  
[nltk_data] Downloading package punkt to /root/nltk_data...  
[nltk_data]   Unzipping tokenizers/punkt.zip.  
[nltk_data] Downloading package stopwords to /root/nltk_data...  
[nltk_data]   Unzipping corpora/stopwords.zip.  
[nltk_data] Downloading package wordnet to /root/nltk_data...  
+-----+  
|word |  
+-----+  
|hello |  
|hate |  
|hate |  
|love |  
|dont |  
|want |  
|cant |  
|put |  
|nobody|  
|else |  
+-----+
```